Hanbin Zheng

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Education

Zhejiang University - University of Illinois Urbana-Champaign Institute (Dual degree program):

University of Illinois Urbana-Champaign, Illinois, US — Sep 2024 – Jun 2028
BS in Computer Engineering, GPA: 3.985/4.0

Zhejiang University, Zhejiang, China — Sep 2024 – Jun 2028
BS in Electronic and Computer Engineering, GPA: 3.899/4.0

Research Experience

Undergraduate Researcher, ZJUI — Jun 2025 – Dec 2025
Advisor: **Prof. Xiangming Meng**

- **Flow Models Exploration** GitHub
  Conducted an in-depth study of Flow Matching generative modeling, implementing conditional image generation on MNIST and CIFAR-10 from scratch.
- **MeanFlow for Image Inverse Problem** GitHub
  Investigated one-step generative modeling with the recently proposed MeanFlow models, altered its initial distribution from Gaussian distributions to noisy images, achieving efficient one-step denoising.
- **GMFlow for Inverse Problem** GitHub
  Investigated Variational Flow Matching and Gaussian Mixture Flow Matching for image inverse problem; implemented 2D toy examples to validate our new algorithm.

Selected Projects

RoboMaster Hero Robot Electrical Control System GitHub — Nov 2025 – Apr 2026

- **Embedded Control:** Developed real-time control system on STM32H723 for a steering wheel hero robot, managing high-frequency control loops and peripheral hardware integration.
- **Orientation Estimation:** Independently derived **quaternion kinematics** and implemented the **Mahony Filter** from scratch for IMU sensor fusion, achieving stable orientation tracking.

Deep Reinforcement Learning GitHub — Jul 2026 – Present

- **Curriculum & Implementation:** Self-studied UC Berkeley CS285 (Deep RL), implementing core algorithms from scratch including Behavioral Cloning (Flow Matching Policy), Policy Gradients, Actor-Critic methods, and Deep Q-Learning.
- **Experimental Evaluation:** Benchmarked agent performance and trained policies across diverse environments such as PushT, InvertedPendulum, and LunarLander.

Skills

Programming Languages: Python (PyTorch), C/C++, LaTeX

Development Tools: Git, Linux, Makefile, CMake, Vim/NeoVim, Emacs

Deep Learning: Unet, Transformer, ViT, DiT

Generative Modeling: Flow Matching, MeanFlow, Diffusion, GMFlow

Robotics & Embedded: STM32, Basic ROS2/ROS2 Control, Quaternion & Euler Angles, Soldering

Reinforcement Learning: Behavioral Cloning, Model-Free RL (Policy Gradient, Actor-Critic, DQN)

Awards and Honors

First Prize in ZJU Mathematical Modeling Competition — Jun 2025
Zhejiang Provincial Scholarship — Oct 2025
Zhejiang University Second Scholarship — Oct 2025
Third Prize in National RoboMaster University League (Shanghai, 3v3) — Apr 2026